Derivations and numerical checks for optimal focusing

Companion notes to Secs. 3.1–3.2 of "Fundamental limits in photonics and electromagnetics: a tutorial"

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Abstract

These notes collect the derivations behind the free-space focusing results (tutorial Sec. 3.1) and the complex-media focusing results (Sec. 3.2) of the tutorial by Miller and Monticone, at a level of detail that would be out of place in a tutorial of that breadth. Every closed-form result is checked numerically by a script in the accompanying repository. Each check is independent of the derivation it tests, e.g. brute-force quadrature with explicit polarization vectors, or a full multiple-scattering model. The achieved agreement is quoted in green alongside each result. Several results here are new relative to the tutorial text: closed forms for the full (anisotropic) focal-plane fields, the exact energy penalties for super-oscillatory spots, a closed form for the three-dimensional space-time focus, and explicit symmetry conditions for the point-dipole media used to test Sec. 3.2.

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1 Conventions

Fields are complex amplitudes with time dependence $e^{-i\omega t}$, the time-averaged Poynting vector is $\frac{1}{2}\operatorname{Re}(\mathbf{E}\times\mathbf{H}^*)$, and $k=\omega/c$. The electric energy density at a point is $U=\frac{1}{4}\varepsilon_0|\mathbf{E}|^2$, as in the tutorial. The numerical codes use natural units $\varepsilon_0=\mu_0=c=Z_0=1$ and report dimensionless ratios such as $(2\pi c/k^2)\,\varepsilon_0|\mathbf{E}(0)|^2/P$.

Spherical Bessel functions are denoted j_ℓ, cylindrical ones J_n, and $\mathrm{Si}(u)=\int_0^u \sin t/t\,\mathrm{d}t$. Two integral identities are used repeatedly:

$$\int_{-\pi}^{\pi} e^{iu\cos\varphi}\cos(n\varphi)\,\mathrm{d}\varphi = 2\pi i^n J_n(u), \qquad \int_{-\pi}^{\pi} e^{iu\cos\varphi}\sin(n\varphi)\,\mathrm{d}\varphi = 0, \tag{1}$$

$$\int_0^1 x^{\nu+1}(1-x^2)^{\mu}J_\nu(ax)\,\mathrm{d}x = 2^\mu\Gamma(\mu+1)\,a^{-\mu-1}J_{\nu+\mu+1}(a) \qquad \text{(Sonine)}, \tag{2}$$

the first being the Jacobi–Anger ("Hansen–Bessel") identity, and the second Sonine's first finite integral ($\operatorname{Re}\nu>-1$, $\operatorname{Re}\mu>-1$). With $j_\ell(a)=\sqrt{\pi/(2a)}\,J_{\ell+1/2}(a)$, the special cases needed below are

$$\begin{aligned}
\int_0^1 \frac{\beta J_0(a\beta)}{\sqrt{1-\beta^2}}\mathrm{d}\beta &= j_0(a), & \int_0^1 \beta\sqrt{1-\beta^2}\,J_0(a\beta)\mathrm{d}\beta &= \frac{j_1(a)}{a},\\
\int_0^1 \frac{\beta^3 J_2(a\beta)}{\sqrt{1-\beta^2}}\mathrm{d}\beta &= j_2(a), & \int_0^1 \beta^2\sqrt{1-\beta^2}\,J_1(a\beta)\mathrm{d}\beta &= \frac{j_2(a)}{a},
\end{aligned} \tag{3}$$

together with the elementary results $\int_0^s \beta J_0(a\beta)\mathrm{d}\beta = sJ_1(as)/a$, $\int_0^s \beta^2 J_1(a\beta)\mathrm{d}\beta = s^2J_2(as)/a$, and $\int_0^s \beta^3 J_2(a\beta)\mathrm{d}\beta = s^3J_3(as)/a$ (all from $\frac{\mathrm{d}}{\mathrm{d}x}[x^nJ_n(x)]=x^nJ_{n-1}(x)$), and $\int_0^s \beta^3 J_0(a\beta)\mathrm{d}\beta = s^3J_1(as)/a - 2s^2J_2(as)/a^2$.

The extremal-eigenvalue lemma. Every bound in these notes is an instance of the following. For a linear functional $f(c)=\langle v,c\rangle$ and a positive weight g,

$$\frac{|\langle v,c\rangle|^2}{\langle c, g\,c\rangle} \le \langle v, g^{-1}v\rangle, \qquad \text{with equality iff } c\propto g^{-1}v, \tag{4}$$

by the Cauchy–Schwarz inequality applied to $g^{1/2}c$ and $g^{-1/2}v$. Equivalently, the quadratic form $c^\dagger(vv^\dagger)c$ is rank one, so its Rayleigh quotient is maximized by its only nonzero eigenvector.

2 Focusing from a cone of plane waves (tutorial Sec. 3.1.2)

2.1 Plane-wave expansion and power

A field produced by sources in $z<0$ and observed in a region free of sources is a superposition of propagating plane waves (evanescent components are excluded, as they are absent at far-field

distances from the sources):

$$\mathbf{E}(\mathbf{x}) = \sum_p \int \mathrm{d}^2\mathbf{k}_\parallel \, c_p(\mathbf{k}_\parallel)\, \hat{\mathbf{e}}_p(\mathbf{k}_\parallel)\, e^{i\mathbf{k}\cdot\mathbf{x}}, \qquad \mathbf{H}(\mathbf{x}) = \frac{1}{Z_0} \sum_p \int \mathrm{d}^2\mathbf{k}_\parallel \, c_p(\mathbf{k}_\parallel)\, \hat{\mathbf{k}} \times \hat{\mathbf{e}}_p(\mathbf{k}_\parallel)\, e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{5}$$

with $\mathbf{k} = \mathbf{k}_\parallel + k_z\hat{\mathbf{z}}$, $k_z = \sqrt{k^2 - |\mathbf{k}_\parallel|^2} \geq 0$, and $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{k}})$ a right-handed orthonormal triad. Numerical aperture $\sin\theta$ restricts $|\mathbf{k}_\parallel| \leq k \sin\theta$. The z-directed power through a plane is

$$\begin{aligned} P &= \frac{1}{2}\,\mathrm{Re}\int (\mathbf{E}\times\mathbf{H}^*)\cdot\hat{\mathbf{z}}\,\mathrm{d}x\,\mathrm{d}y = \frac{(2\pi)^2}{2Z_0}\,\mathrm{Re}\sum_{p,p'}\int \mathrm{d}^2\mathbf{k}_\parallel\, c_p c_{p'}^*\,[\hat{\mathbf{e}}_p \times (\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{p'}^*)]\cdot\hat{\mathbf{z}} \\ &= \frac{2\pi^2}{Z_0}\sum_p \int \mathrm{d}^2\mathbf{k}_\parallel\, \frac{k_z}{k}|c_p(\mathbf{k}_\parallel)|^2, \end{aligned} \tag{6}$$

where the spatial integral produced $(2\pi)^2\delta(\mathbf{k}_\parallel - \mathbf{k}'_\parallel)$ and $\hat{\mathbf{e}}_p\times(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{p'}^*) = \hat{\mathbf{k}}(\hat{\mathbf{e}}_p\cdot\hat{\mathbf{e}}_{p'}^*) - \hat{\mathbf{e}}_{p'}^*(\hat{\mathbf{e}}_p\cdot\hat{\mathbf{k}}) = \hat{\mathbf{k}}\,\delta_{pp'}$. [numerical check: `power_normalization_check.py`: direct Poynting integration of FFT-synthesized fields with random complex spectra in both polarizations, relative error $\leq 7\times10^{-16}$]

2.2 Optimal coefficients and the focal matrix

The field at the focus in a (possibly complex) polarization $\hat{\mathbf{q}}$ is $\hat{\mathbf{q}}^*\cdot\mathbf{E}(0) = \sum_p \int \mathrm{d}^2\mathbf{k}_\parallel\, c_p\,(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)$. We consider two normalizations,

$$\text{power:}\quad \frac{2\pi^2}{Z_0}\sum_p\int \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k_z}{k}|c_p|^2 = P, \qquad\qquad \text{amplitude:}\quad \frac{2\pi^2}{Z_0}\sum_p\int \mathrm{d}^2\mathbf{k}_\parallel\,|c_p|^2 \equiv P_{\mathrm{amp}}, \tag{7}$$

i.e. weights $g = k_z/k$ and $g = 1$ in (4). The optimal coefficients are $c_p \propto g^{-1}(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)^*$, i.e. $c_p^{(\mathrm{opt})} \propto (k/k_z)(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)^*$ for power normalization, which is the tutorial's $\alpha_p^{(\mathrm{opt})} \propto \sqrt{k/k_z}(\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p)^*$ written for $\alpha_p = \sqrt{k_z/k}\,c_p$. The maximum is

$$\begin{aligned} \max\frac{\varepsilon_0|\hat{\mathbf{q}}^*\cdot\mathbf{E}(0)|^2}{P} &= \frac{\varepsilon_0 Z_0}{2\pi^2}\sum_p\int \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k}{k_z}|\hat{\mathbf{q}}^*\cdot\hat{\mathbf{e}}_p|^2 = \frac{1}{2\pi^2 c}\,\hat{\mathbf{q}}^\dagger\mathsf{M}\,\hat{\mathbf{q}}, \\ \mathsf{M} &= \int \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k}{k_z}(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}}), \end{aligned} \tag{8}$$

using the completeness relation $\sum_p \hat{\mathbf{e}}_p\hat{\mathbf{e}}_p^\dagger = \mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}}$. For amplitude normalization the factor k/k_z is absent.

Solid-angle form. Since $|\mathbf{k}_\parallel| = k\sin\theta'$ and $\mathrm{d}|\mathbf{k}_\parallel| = k\cos\theta'\mathrm{d}\theta'$,

$$\mathrm{d}^2\mathbf{k}_\parallel = k^2\cos\theta'\;\mathrm{d}\Omega, \qquad \text{so} \qquad \mathrm{d}^2\mathbf{k}_\parallel\,\frac{k}{k_z} = k^2\;\mathrm{d}\Omega. \tag{9}$$

Power normalization therefore weights all directions *uniformly in solid angle*. This observation organizes everything that follows. In particular, the power-optimal field for target $\hat{\mathbf{q}}$ is

$$\mathbf{E}^{(\mathrm{opt})}(\mathbf{x}) \propto k^2\int_{\mathrm{cap}} \mathrm{d}\Omega\,(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{q}}\,e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}}, \tag{10}$$

a Herglotz wave with uniform density over the cap of allowed directions. The amplitude-optimal field has an extra factor $\cos\theta'$ in the integrand.

2.3 Focal-point intensity maxima

The azimuthal integrals of the off-diagonal entries of M ($\propto \cos\varphi' \sin\varphi'$, $\cos\varphi'$, $\sin\varphi'$) vanish, so M is diagonal with $\mathsf{M}_{xx} = \mathsf{M}_{yy}$, and

$$\begin{aligned}\mathsf{M}_{xx} &= 2\pi k^2 \int_0^\theta \sin\theta' \Big(1 - \tfrac{1}{2}\sin^2\theta'\Big)\mathrm{d}\theta' = \pi k^2 \Big[\tfrac{4}{3} - \cos\theta - \tfrac{1}{3}\cos^3\theta\Big],\\ \mathsf{M}_{zz} &= 2\pi k^2 \int_0^\theta \sin^3\theta' \mathrm{d}\theta' = 2\pi k^2 \Big[\tfrac{2}{3} - \cos\theta + \tfrac{1}{3}\cos^3\theta\Big].\end{aligned} \tag{11}$$

Since $\hat{\mathbf{q}}^\dagger \mathsf{M} \hat{\mathbf{q}} \le \lambda_{\max}(\mathsf{M})$ for any unit $\hat{\mathbf{q}}$, including complex (e.g. circular) polarizations, the global optimum is in-plane polarization, because

$$\mathsf{M}_{xx} - \mathsf{M}_{zz} = \pi k^2 \cos\theta \sin^2\theta \ \ge 0. \tag{12}$$

Collecting results in units of $k^2/(2\pi c)$, with $P \to P_{\mathrm{amp}}$ for the amplitude-normalized entries:

$$\frac{2\pi c}{k^2}\frac{\varepsilon_0 |\mathbf{E}(0)|^2}{P} \le \begin{cases} \frac{4}{3} - \cos\theta - \frac{1}{3}\cos^3\theta & \text{power, in-plane } \hat{\mathbf{q}} \text{ (global max.)} \\ 2\big(\frac{2}{3} - \cos\theta + \frac{1}{3}\cos^3\theta\big) & \text{power, } \hat{\mathbf{q}} = \hat{\mathbf{z}} \\ \sin^2\theta\,\big(1 - \frac{1}{4}\sin^2\theta\big) & \text{amplitude, in-plane} \\ \frac{1}{2}\sin^4\theta & \text{amplitude, } \hat{\mathbf{q}} = \hat{\mathbf{z}}, \end{cases} \tag{13}$$

[numerical check: `focal_intensity_bounds.py`: the 3×3 matrix built with *explicit* TE/TM vectors reproduces all four entries to 3×10^{-13}, off-diagonals $< 10^{-16}$; an independent Monte-Carlo discretization with random polarization bases agrees to its sampling error]

Limits. Paraxially all in-plane entries tend to θ^2 and the z entries vanish as $\theta^4/2$. At 2π solid angle the power-normalized maximum is $4/3$ for *every* polarization, giving $U(\mathbf{x}_0)/P \le k^2/(6\pi c)$. The amplitude-normalized values are $3/4$ (in-plane) and $1/2$ (z), so the polarization-independence at 2π holds for power normalization only.

Amplitude normalization versus physical power. The amplitude-normalized entries of (13) are intensities per unit P_{amp}, not per unit physical power. The ratio $\frac{4/3}{3/4} = 16/9 \approx 1.78$ between the two maxima therefore compares different denominators. The amplitude-optimal beam, $c_p = (\hat{\mathbf{x}}\cdot\hat{\mathbf{e}}_p)^*$, measured against its physical power (6), delivers

$$\frac{2\pi c}{k^2}\frac{\varepsilon_0 |\mathbf{E}(0)|^2}{P} = \frac{\sin^4\theta\,(1 - \frac{1}{4}\sin^2\theta)^2}{2(\frac{4}{15} - \frac{1}{6}\cos^3\theta - \frac{1}{10}\cos^5\theta)} \xrightarrow{\theta\to\pi/2} \frac{135}{128} \approx 1.055, \tag{14}$$

where the denominator follows from

$$\int \mathrm{d}^2\mathbf{k}_\parallel \frac{k_z}{k}\Big(1 - \frac{k_x^2}{k^2}\Big) = 2\pi k^2 \int_{\cos\theta}^1 u^2 \Big(\frac{1}{2} + \frac{u^2}{2}\Big)\mathrm{d}u.$$

At equal physical power, then, the power-optimal beam exceeds the amplitude-optimal one by $(4/3)/(135/128) = 1.264$ at 2π solid angle. [numerical check: `focal_intensity_bounds.py`: explicit synthesis of the amplitude-optimal beam, rel. error $< 10^{-14}$]

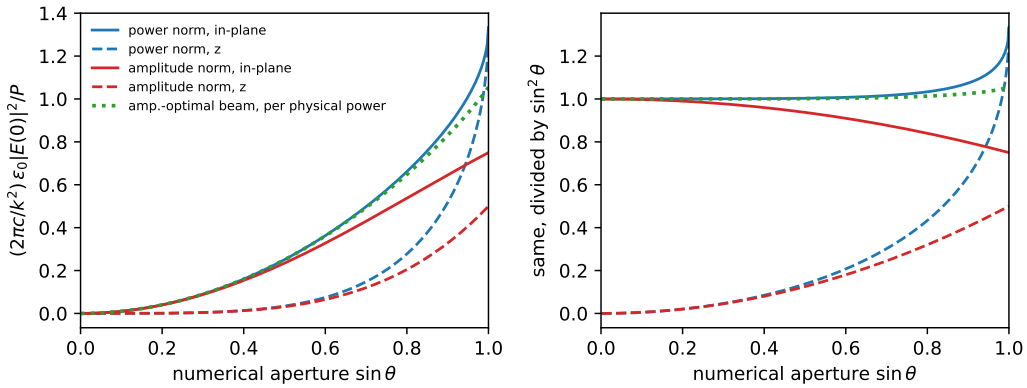


Figure 1: Maximum focal intensity versus numerical aperture, (13). The dotted curve is the amplitude-optimal beam measured per unit physical power, (14).

Reference beams and comparison with Dias & García de Abajo. For a family of reference beams $c_p \propto (k_z/k)^n(\hat{\mathbf{x}}\cdot\hat{\mathbf{e}}_p)^*$, the same algebra gives, at 2π solid angle and per unit physical power,

$$\frac{2\pi c}{k^2}\frac{\varepsilon_0|\mathbf{E}(0)|^2}{P} = \frac{\left[\frac{1}{n+2}+\frac{1}{n+4}\right]^2}{\frac{1}{2n+3}+\frac{1}{2n+5}} = \begin{cases} 4/3 & n=-1 \text{ (power-optimal)} \\ 1.2094 & n=-\frac{1}{2} \text{ (aplanatic lens, uniform pupil)} \\ 135/128 & n=0 \text{ (amplitude-optimal)} \\ 0.9292 & n=\frac{1}{2}. \end{cases} \tag{15}$$

The last case is the "diffraction-limited Gaussian" reference of Dias and García de Abajo (their $\beta \propto \hat{\mathbf{n}}\cdot\hat{\mathbf{e}}\sqrt{k_z/k}$). The optimum exceeds it by 1.435, the factor ≈ 1.43 quoted in the tutorial. Against an aplanatic lens with uniformly filled pupil, $c \propto (k/k_z)^{1/2}$, the advantage is only 1.10. Their focal-intensity result, Eq. (4) of that paper, $\max|\mathbf{E}(0)|^2/P_{\rm inc} = 4\omega^2/3c^3$ for 4π in Gaussian units with $\mathbf{E}_{\rm phys} = 2\,\mathrm{Re}[\mathbf{E}e^{-i\omega t}]$, agrees exactly with $(2\pi c/k^2)\varepsilon_0|\mathbf{E}|^2/P = 8/3$ derived here: converting to Gaussian units gives $|\mathbf{E}|^2/P = 16\omega^2/3c^3$, and halving the amplitude convention divides this by 4. The time-averaged energy density in their convention is $(|\mathbf{E}|^2+|\mathbf{H}|^2)/4\pi$, not $/8\pi$, which accounts for the factor of 2 between their energy-density bound and that of Bassett (and (13)). [numerical check: `focal_intensity_bounds.py`: closed form vs. explicit synthesis for $n=-1,-\frac{1}{2},0,\frac{1}{2},1$, agreement to 6 digits]

3 The optimal fields across the focal plane

3.1 General numerical aperture

Take $\hat{\mathbf{q}} = \hat{\mathbf{x}}$, write $\hat{\mathbf{k}} = (\sin\theta'\cos\varphi', \sin\theta'\sin\varphi', \cos\theta')$ and let the field point be at cylindrical coordinates (ρ,ϕ,z), with $a = k\rho$. The Cartesian components of $(\mathbb{I}-\hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{x}}$ are

$$(1-\sin^2\theta'\cos^2\varphi',\ -\tfrac{1}{2}\sin^2\theta'\sin 2\varphi',\ -\sin\theta'\cos\theta'\cos\varphi').$$

Shifting $\varphi' \to \varphi'+\phi$ in (1) gives

$$\begin{aligned} \int \mathrm{d}\varphi'\, e^{iu\cos(\varphi'-\phi)}\{1,\ \cos^2\varphi'\} &= \{2\pi J_0(u),\ \pi[J_0(u)-\cos 2\phi\, J_2(u)]\}, \\ \int \mathrm{d}\varphi'\, e^{iu\cos(\varphi'-\phi)}\{\sin 2\varphi',\ \cos\varphi'\} &= \{-2\pi\sin 2\phi\, J_2(u),\ 2\pi i\cos\phi\, J_1(u)\}, \end{aligned} \tag{16}$$

with $u = a\sin\theta'$. Hence, with $w(\theta') = 1$ (power) or $|\cos\theta'|$ (amplitude normalization),

$$\begin{aligned} E_x &= \pi k^2 \int \mathrm{d}\theta'\ \sin\theta'\, w\, e^{ikz\cos\theta'} \Big[(2-\sin^2\theta')J_0(u) + \sin^2\theta'\cos 2\phi\, J_2(u)\Big], \\ E_y &= \pi k^2 \sin 2\phi \int \mathrm{d}\theta'\ \sin^3\theta'\, w\, e^{ikz\cos\theta'} J_2(u), \\ E_z &= -2\pi i k^2 \cos\phi \int \mathrm{d}\theta'\ \sin^2\theta'\cos\theta'\, w\, e^{ikz\cos\theta'} J_1(u), \end{aligned} \tag{17}$$

with $\theta' \in [0,\theta]$ for one-sided excitation, and in addition $\theta' \in [\pi-\theta,\pi]$ for two-sided excitation. The focal spot is *not* rotationally symmetric: E_x depends on $\cos 2\phi$, and $E_y \propto \sin 2\phi$ and $E_z \propto \cos\phi$ both vanish on the axes. In particular E_z is antisymmetric in x, with two lobes of opposite sign on either side of the focus along the polarization axis. [numerical check: `focal_plane_fields.py`: (17) vs. brute-force 2D quadrature with explicit polarization vectors, rel. error 2×10^{-14}]

3.2 Closed forms at 2π solid angle

For the power-optimal beam at $\theta = \pi/2$ and $z = 0$, substitute $\beta = \sin\theta'$, so that $\sin\theta'\mathrm{d}\theta' = \beta\,\mathrm{d}\beta/\sqrt{1-\beta^2}$. The integrals reduce to the cases (3), together with

$$\int_0^1 \frac{\beta^3 J_0(a\beta)}{\sqrt{1-\beta^2}}\mathrm{d}\beta = j_0(a) - \int_0^1 \beta\sqrt{1-\beta^2}J_0(a\beta)\mathrm{d}\beta = j_0(a) - \frac{j_1(a)}{a}.$$

With $j_2 = 3j_1/a - j_0$:

$$\begin{aligned} E_x &= 2\pi k^2\Big[\sin^2\phi\, j_0(a) + (3\cos^2\phi - 1)\frac{j_1(a)}{a}\Big], \\ E_y &= 2\pi k^2 \sin\phi\cos\phi\, j_2(a), \qquad E_z = -2\pi i k^2\cos\phi\,\frac{J_2(a)}{a}, \end{aligned} \tag{18}$$

with $E_x(0) = 4\pi k^2/3$. Comparing with the full-sphere result (22) below shows that the in-plane part of (18) is *exactly one half* of the 4π-optimal field in the focal plane. This follows from symmetry: under $z \to -z$ the lower cap maps to the upper one, the in-plane components of $(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{x}}$ are even in $\hat{k}_z$, and E_z is odd. Adding the mirror hemisphere therefore doubles $\mathbf{E}_\parallel$ and cancels E_z at $z = 0$. [numerical check: closed form vs. quadrature 1.6×10^{-14}; $|\mathbf{E}_\parallel^{2\pi} - \tfrac{1}{2}\mathbf{E}_\parallel^{4\pi}|/|\mathbf{E}| < 10^{-16}$]

Along the polarization axis ($\phi = 0$) and perpendicular to it ($\phi = \pi/2$):

$$|\mathbf{E}|^2_{\phi=0} = 4\pi^2k^4\Big[4\Big(\frac{j_1(a)}{a}\Big)^2 + \Big(\frac{J_2(a)}{a}\Big)^2\Big], \qquad |\mathbf{E}|^2_{\phi=\pi/2} = 4\pi^2k^4\big[j_0(a) - j_1(a)/a\big]^2. \tag{19}$$

The first is the expression printed in the tutorial, which is therefore the cut along the polarization axis. The second has *exactly the same shape* as the 4π profile in the plane perpendicular to the polarization (Sec. 4). The 2π-optimal spot is elliptical: its width along the polarization axis exceeds the paraxial Airy width, while its width perpendicular to the polarization is already the 4π value (Table 1).

Target $\hat{\mathbf{q}} = \hat{\mathbf{z}}$. The same steps with $(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{z}} = (-\sin\theta'\cos\theta'\cos\varphi', -\sin\theta'\cos\theta'\sin\varphi', \sin^2\theta')$ give a rotationally symmetric focal-plane field,

$$E_z = 2\pi k^2\Big[j_0(a) - \frac{j_1(a)}{a}\Big], \qquad E_\rho = -2\pi i k^2\,\frac{J_2(a)}{a}, \tag{20}$$

so $|\mathbf{E}|^2 \propto (j_0 - j_1/a)^2 + (J_2/a)^2$. The radial component is odd in $\hat{k}_z$ and is cancelled by the second hemisphere. This is why only 4π excitation gives a purely z-polarized focal plane, as stated in the tutorial. [numerical check: brute force vs. closed form, 2×10^{-14}]

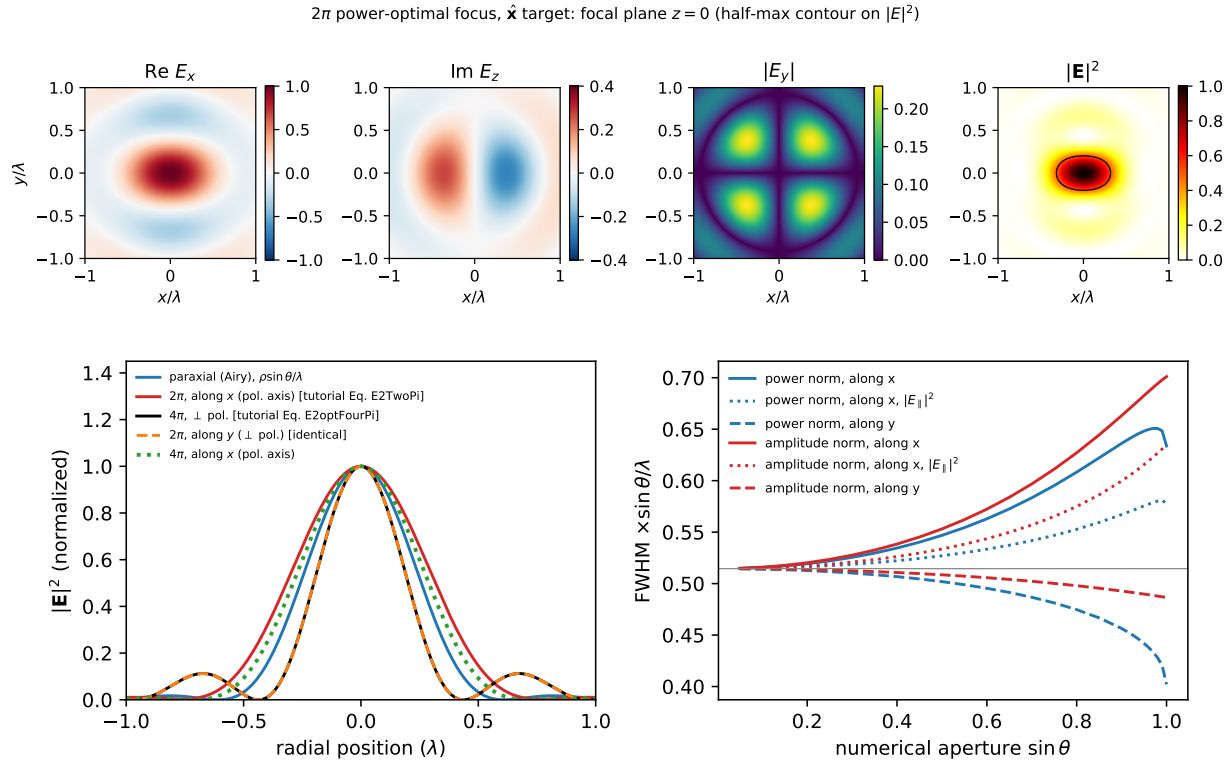


Figure 2: Top: focal-plane fields of the 2π power-optimal beam, (18). $E_z \propto \cos\phi$ is antisymmetric in x, and the half-maximum contour of $|\mathbf{E}|^2$ is an ellipse elongated along the polarization. Bottom left: intensity cuts. The 2π profile perpendicular to the polarization (dashed orange) coincides with the 4π profile (black). Bottom right: FWHM versus NA along the polarization axis (x, solid) and perpendicular to it (y, dashed); dotted curves show the in-plane intensity $|\mathbf{E}_\parallel|^2$ along x.

Amplitude normalization, any NA. The extra $\cos\theta'$ turns the integrals into elementary Bessel integrals ($s = \sin\theta$, Bessel arguments as):

$$E_x = \pi k^2 \Big[(2s - s^3)\frac{J_1}{a} + 2s^2\frac{J_2}{a^2} + \cos 2\phi\, s^3\frac{J_3}{a}\Big], \qquad E_y = \pi k^2 \sin 2\phi\, s^3\frac{J_3}{a}, \tag{21}$$

and at $s = 1$, $E_z = -2\pi i k^2 \cos\phi\, j_2(a)/a$ [last case of (3)]. At $\phi = 0$ the recurrences reduce E_x to $\pi k^2 s^2[(1 - \frac{s^2}{4})J_0 + J_2 + \frac{s^2}{4}J_4]$. [numerical check: `tests/test_sec3_1.py`: rel. error $< 5 \times 10^{-14}$ at $s = 0.3, 0.8, 1$]

Paraxial limit. For $\theta \ll 1$ both normalizations give the Airy pattern,

$$E_x \simeq \int_{|\mathbf{k}_\parallel| \le k\sin\theta} \mathrm{d}^2\mathbf{k}_\parallel\, e^{i\mathbf{k}_\parallel\cdot\mathbf{x}_\parallel} = 2\pi k^2 \sin^2\theta\, \frac{J_1(a\sin\theta)}{a\sin\theta}, \qquad E_y, E_z \to 0.$$

Two features of Table 1 and Fig. 2 deserve emphasis:

1. In normalized units ($\rho\sin\theta/\lambda$), increasing the NA *broadens* the spot along the polarization axis, from 0.51 to 0.63, but *narrows* it perpendicular to the polarization, from 0.51 to 0.40. The width along x is non-monotonic very close to 2π (maximum 0.651 near $\sin\theta \approx 0.975$); this is a real feature of the optimal field, not a numerical artifact.

Table 1: Full widths at half maximum of $|\mathbf{E}|^2$ and locations of the first minimum, in wavelengths (paraxial row in units of $\lambda/\sin\theta$). "First min." is a true zero only where $I/I_0 \approx 0$. The 4π beam is isotropic in the plane perpendicular to its polarization, so its y and z rows coincide.

excitation	direction	FWHM	first min.	I/I_0 there
paraxial (Airy)	any	0.5145	0.6098	0
2π, power norm.	x (polarization)	0.6337	0.7650	2.8×10^{-3}
	y	0.4021	0.4367	0
	z (optical axis)	0.8867	0.9898	1.6×10^{-2}
2π, amplitude norm.	x	0.7011	0.8734	1.4×10^{-3}
	y	0.4865	0.5600	0
	z	1.1862	–	–
4π (power norm.)	x (polarization)	0.5777	0.7151	0
	y, z	0.4021	0.4367	0

2. Relative to 2π, the 4π optimum is tighter in the focal plane only along the polarization axis, because the lower hemisphere cancels E_z (0.634 to 0.578). The large gain is along the optical axis (0.887 to 0.402). Perpendicular to the polarization the two profiles are identical.

4 Full 4π solid angle (tutorial Sec. 3.1.3)

4.1 Herglotz form and the imaginary part of the Green's function

For the full sphere, (10) can be evaluated in closed form from

$$\int \mathrm{d}\Omega\, e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}} = 4\pi j_0(kr), \qquad \int \mathrm{d}\Omega\, \hat{k}_i\hat{k}_j e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}} = -k^{-2}\partial_i\partial_j[4\pi j_0(kr)].$$

Using $j_0' = -j_1$ and $j_1' = j_0 - 2j_1/x$ ($x = kr$):

$$k^2\int \mathrm{d}\Omega\,(\mathbb{I}-\hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}} = 4\pi k^2\Big[\Big(j_0-\frac{j_1}{x}\Big)\mathbb{I}+\Big(\frac{3j_1}{x}-j_0\Big)\hat{\mathbf{r}}\hat{\mathbf{r}}^{\mathrm{T}}\Big] = 16\pi^2 k\ \mathrm{Im}\,\mathsf{G}_0(\mathbf{x},0), \tag{22}$$

where $\mathsf{G}_0 = (\mathbb{I}+\nabla\nabla/k^2)e^{ikr}/(4\pi r)$ is the free-space dyadic Green's function. [numerical check: `fourpi_spherical_waves.py`: 8×10^{-15}] The 4π-optimal field is thus $\mathrm{Im}\,\mathsf{G}_0$ (the time-reversal field of Sec. 7.4), and the 2π-optimal field is "half of $\mathrm{Im}\,\mathsf{G}_0$", restricted to one hemisphere of directions. In the plane perpendicular to $\hat{\mathbf{q}}$ ($\hat{\mathbf{r}}\cdot\hat{\mathbf{q}} = 0$) the field is $4\pi k^2(j_0 - j_1/x)\hat{\mathbf{q}}$, isotropic in that plane. Along $\hat{\mathbf{q}}$ it is $8\pi k^2(j_1/x)\hat{\mathbf{q}}$.

4.2 Vector spherical waves

Define normalized vector spherical waves $\mathbf{M}_{\ell m} = z_\ell(kr)\mathbf{X}_{\ell m}$ and $\mathbf{N}_{\ell m} = k^{-1}\nabla\times\mathbf{M}_{\ell m}$, with $\mathbf{X}_{\ell m} = \mathbf{L}Y_{\ell m}/\sqrt{\ell(\ell+1)}$. Explicitly,

$$\mathbf{N}_{\ell m} = i\left[\sqrt{\ell(\ell+1)}\,\frac{z_\ell(x)}{x}Y_{\ell m}\hat{\mathbf{r}} + \frac{1}{\sqrt{\ell(\ell+1)}}\,\frac{1}{x}\,\frac{\mathrm{d}[xz_\ell(x)]}{\mathrm{d}x}\nabla_s Y_{\ell m}\right], \tag{23}$$

with $\nabla_s = \hat{\boldsymbol{\theta}}\partial_\theta + \hat{\boldsymbol{\phi}}\sin^{-1}\theta\,\partial_\phi$ and $z_\ell \in \{j_\ell, h_\ell^{(1)}, h_\ell^{(2)}\}$. [numerical check: finite-difference $\nabla\times\mathbf{M}$ vs. $k\mathbf{N}$: 2×10^{-10}] For regular waves at $r\to0$: $\mathbf{M}_{\ell m}\propto j_\ell\to0$; the radial part of $\mathbf{N}_{\ell m}$ is $\propto j_\ell(x)/x$,

and the tangential part $\propto x^{-1}\mathrm{d}(xj_\ell)/\mathrm{d}x \to (\ell+1)x^{\ell-1}/(2\ell+1)!!$. Both vanish unless $\ell = 1$. At the origin only the three electric-dipole waves survive, and

$$\mathbf{N}_{1,0}(0) = i\,\tfrac{1}{\sqrt{2}}\,\tfrac{2}{3}\sqrt{\tfrac{3}{4\pi}}\,(\cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}) = i\,\tfrac{2}{3}\sqrt{\tfrac{3}{8\pi}}\,\hat{\mathbf{z}}, \tag{24}$$

and similarly for $m = \pm 1$ along the spherical basis vectors. [numerical check: scan of all regular VSWs with $\ell \le 6$: only $(1, m)$ nonzero; $|\mathbf{N}_{1m}(0)| = 0.2303294330$ exactly] In the equatorial plane, $\mathrm{d}(xj_1)/\mathrm{d}x = xj_0 - j_1$ gives $\mathbf{N}_{1,0} = -i\sqrt{3/(8\pi)}\,(j_0 - j_1/x)\,\hat{\boldsymbol{\theta}}$, which is the tutorial's equatorial expression up to the overall phase convention.

Incoming power and the 4π bound. The regular wave splits as $\mathbf{N}^{\mathrm{reg}} = \frac{1}{2}(\mathbf{N}^{(1)} + \mathbf{N}^{(2)})$. In the far field $h_\ell^{(2)}(x) \to i^{\ell+1}e^{-ix}/x$, $\mathbf{N}^{(2)}$ is transverse with $\int |\mathbf{N}^{(2)}|^2 r^2\,\mathrm{d}\Omega \to k^{-2}\int |\nabla_s Y|^2\,\mathrm{d}\Omega/[\ell(\ell+1)] = k^{-2}$, and $|\mathbf{H}| = |\mathbf{E}|/Z_0$. Hence the incoming part $\frac{1}{2}\mathbf{N}_{1,0}^{(2)}$ carries $P_{\mathrm{in}} = \frac{1}{2}\cdot\frac{1}{4}\cdot\frac{1}{Z_0k^2} = 1/(8Z_0k^2)$, and

$$\frac{U(\mathbf{x}_0)}{P_{\mathrm{in}}} = \frac{\frac{1}{4}\varepsilon_0\cdot\frac{4}{9}\cdot\frac{3}{8\pi}}{1/(8Z_0k^2)} = \frac{k^2}{3\pi c}. \tag{25}$$

[numerical check: inward Poynting flux at $R = 3, 30, 300$: 0.1250000000; $U/P_{\mathrm{in}} = 0.1061032954 = 1/(3\pi)$] The three channels $\mathbf{N}_{1,m}$ have mutually orthogonal values at the origin and orthogonal fluxes. For any target $\hat{\mathbf{q}}$, (4) therefore gives $\max|\hat{\mathbf{q}}^*\cdot\mathbf{E}(0)|^2/P_{\mathrm{in}} = \sum_m |\hat{\mathbf{q}}^*\cdot\mathbf{N}_{1m}(0)|^2/P_1 = |\mathbf{N}_{10}(0)|^2/P_1$, which is independent of $\hat{\mathbf{q}}$.

Doubling, in the plane-wave picture. For a Herglotz wave $\mathbf{E} = \int \mathrm{d}\Omega\,\mathbf{a}(\hat{\mathbf{k}})e^{ik\hat{\mathbf{k}}\cdot\mathbf{x}}$, stationary phase gives $\mathbf{E} \sim (2\pi/ikr)[\mathbf{a}(\hat{\mathbf{r}})e^{ikr} - \mathbf{a}(-\hat{\mathbf{r}})e^{-ikr}]$ as $r\to\infty$, so $P_{\mathrm{in}} = (2\pi^2/Z_0k^2)\int|\mathbf{a}|^2\,\mathrm{d}\Omega$. For a one-sided beam this equals the z-directed power (6), by (9). With $\mathbf{a} = k^2(\mathbb{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^{\mathrm{T}})\hat{\mathbf{q}}$: $|\mathbf{E}(0)|^2 = (8\pi k^2/3)^2$ and $P_{\mathrm{in}} = (2\pi^2/Z_0)(8\pi/3)k^2$ for the full sphere, so $(2\pi c/k^2)\varepsilon_0|\mathbf{E}(0)|^2/P_{\mathrm{in}} = 8/3$. Each hemisphere alone carries half the power and half the field, giving the 2π value $4/3$. The full-sphere Herglotz wave is proportional to $\mathbf{N}_{1,0}^{\mathrm{reg}}$ (ratio constant to 5×10^{-12}), and its incoming power computed from the VSW coefficient agrees with the stationary-phase value, $16\pi^3/3$, to all printed digits.

5 Super-oscillatory spots (tutorial Sec. 3.1.4)

5.1 Minimum energy under point constraints

For $E(x) = \int_{-k_m}^{k_m} c(k)e^{ikx}\mathrm{d}k$ the field energy is $\int|E|^2\mathrm{d}x = 2\pi\int|c|^2\mathrm{d}k$. Minimizing $\int|c|^2$ subject to $E(x_i) = b_i$ is a minimum-norm problem, with solution $c(k) = \sum_i \lambda_i e^{-ikx_i}$ and

$$\min\int|c|^2\mathrm{d}k = \mathbf{b}^\dagger\mathsf{G}^{-1}\mathbf{b}, \qquad \mathsf{G}_{ij} = \int_{-k_m}^{k_m} e^{ik(x_i-x_j)}\mathrm{d}k = \frac{2\sin k_m(x_i-x_j)}{x_i-x_j}. \tag{26}$$

Two points. The single constraint $E(x+\delta) - E(x) = 1$ gives $\min\int|c|^2 = 1/\int|e^{ik\delta}-1|^2\mathrm{d}k = \left[4k_m - 4\sin(k_m\delta)/\delta\right]^{-1} \to 3/(2k_m^3\delta^2)$. This is the tutorial's $E \gtrsim 1/\delta^2$, with its constant.

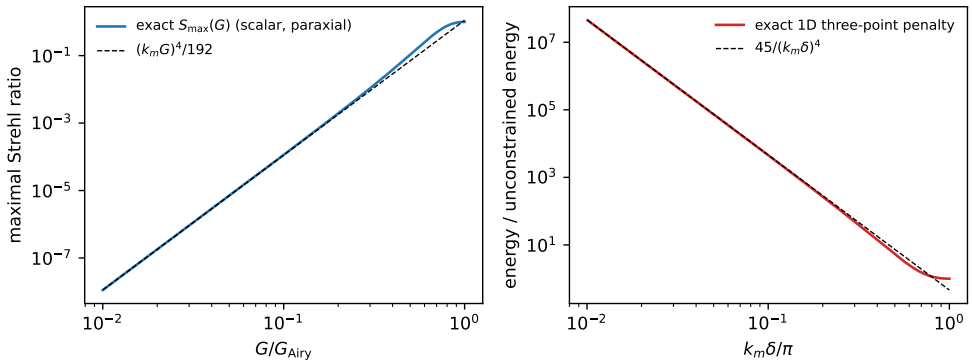


Figure 3: Left: exact maximal Strehl ratio for a zero ring (paraxial, scalar) and its $u^4/192$ asymptote. Right: exact 1D three-point energy penalty and its $45/(k_m\delta)^4$ asymptote.

Three points. Imposing $E(0) = 1$ and $E(\pm\delta) = 0$, the optimum is even in k and the constraints are equivalent to $\int c = 1$ and $\int c\,(1 - \cos k\delta) = 1$. The Gram matrix of $\{1, 1 - \cos k\delta\}$ has entries $2k_m$, $k_m^3\delta^2/3 + O(\delta^4)$ and $k_m^5\delta^4/10 + O(\delta^6)$, so

$$\min \int |c|^2 \mathrm{d}k \;\to\; \frac{2k_m}{2k_m \cdot k_m^5\delta^4/10 - k_m^6\delta^4/9} = \frac{45}{2k_m^5\delta^4}, \qquad \frac{\text{constrained}}{\text{unconstrained}} \to \frac{45}{(k_m\delta)^4}. \tag{27}$$

The unconstrained optimum (c uniform, energy $1/2k_m$) is the sinc beam, whose first zero is at $k_m\delta = \pi$, where the penalty is exactly 1. Halving the spot ($k_m\delta = \pi/2$) costs a factor of 5.28. The asymptotic formula overestimates this, predicting 7.4. [numerical check: `superoscillation.py`: exact vs. asymptote ratio 0.9998 at $k_m\delta = 0.03$]

5.2 Zero ring in two dimensions

In the paraxial scalar model with a circular pupil $|\mathbf{k}_\parallel| \le k_m$, a zero ring at radius G constrains every angular harmonic of the pupil function. The focal amplitude involves only the $m = 0$ harmonic, so the optimum sets the others to zero, leaving the single constraint $\int c_0(k) J_0(kG) k\,\mathrm{d}k = 0$. Projecting the uniform pupil orthogonally to $J_0(kG)$ with the inner product $2\pi \int_0^{k_m} \cdot\, k\,\mathrm{d}k$, and using $\int_0^{k_m} J_0(kG) k\,\mathrm{d}k = k_m J_1(u)/G$ and $\int_0^{k_m} J_0^2(kG) k\,\mathrm{d}k = \frac{k_m^2}{2}[J_0^2(u) + J_1^2(u)]$ with $u = k_m G$, the maximal Strehl ratio is

$$S_{\max}(u) = 1 - \frac{4J_1^2(u)}{u^2[J_0^2(u) + J_1^2(u)]} \xrightarrow{u\to 0} \frac{u^4}{192}, \tag{28}$$

with $S_{\max} = 1$ at the Airy zero $u = 3.8317$. The expansion follows from $[2J_1(u)/u]^2 = 1 - \frac{u^2}{4} + \frac{5u^4}{192}$ and $J_0^2 + J_1^2 = 1 - \frac{u^2}{4} + \frac{6u^4}{192}$. In terms of the spot size normalized to the Airy zero, $S_{\max} \approx 1.12\,(G/G_{\mathrm{Airy}})^4$. The exact penalties are $1/S_{\max} = 9.3$ for a factor-of-2 reduction and 8.7×10^3 for a factor of 10. The naive $2^4 = 16$ therefore overstates the cost of a factor-of-2 reduction by about 1.7×; $S_{\max}(0.5) \approx 0.1$ is also what the $S_{\max}$ curve of the tutorial's super-resolution figure shows. [numerical check: closed form vs. brute-force pupil discretization: $< 10^{-8}$]

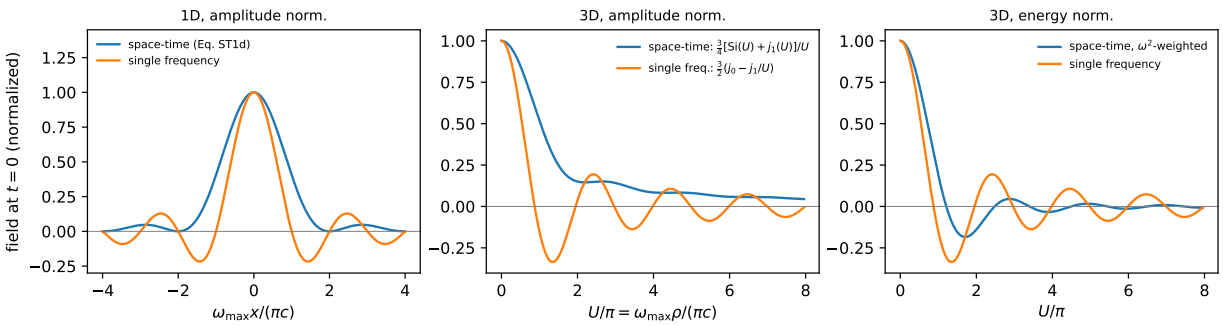


Figure 4: $t = 0$ profiles of space-time-optimal fields (blue) versus single-frequency optima at $\omega_{\max}$ (orange), for the 1D line and 3D, and in 3D for energy normalization.

6 Space-time focusing (tutorial Sec. 3.1.5)

Under amplitude normalization $\int\int |c(\mathbf{k},\omega)|^2 = 1$, (4) makes the optimal c uniform on the support. For the 1D line (transverse wavenumber $|k| \le |\omega|/c$, $|\omega| \le \omega_m$):

$$\psi(x,t) = \int_{-\omega_m}^{\omega_m} \mathrm{d}\omega\, \frac{2\sin(|\omega|x/c)}{x} e^{-i\omega t} = \frac{2}{x}\int_0^{\omega_m} \Big[\sin\omega\Big(\frac{x}{c}+t\Big) + \sin\omega\Big(\frac{x}{c}-t\Big)\Big]\mathrm{d}\omega$$
$$= \frac{4c}{x}\Big[\frac{\sin^2(\omega_m(x+ct)/2c)}{x+ct} + \frac{\sin^2(\omega_m(x-ct)/2c)}{x-ct}\Big], \tag{29}$$

which is the tutorial's result. At $t = 0$ it is $\propto \sin^2(\omega_m x/2c)/x^2$, with first zero at $2\pi c/\omega_m$, twice that of $\sin(\omega_m x/c)/x$. [numerical check: `spacetime_focusing.py`: vs. direct (ω, k) quadrature, 1.5×10^{-14}]

Three dimensions. With uniform weight in ω, the $t = 0$ radial profile is $\int_0^{\omega_m}[j_0 - j_1/u]_{u=\omega\rho/c}\mathrm{d}\omega = (c/\rho)\int_0^U (j_0 - j_1/u)\mathrm{d}u$ with $U = \omega_m\rho/c$. Using $\int_0^U j_1(u)/u\,\mathrm{d}u = \frac{1}{2}[\mathrm{Si}(U) + \cos U/U - \sin U/U^2]$:

$$\psi_{3\mathrm{D}}(\rho,0) \propto \frac{\mathrm{Si}(U) + j_1(U)}{U} = \frac{4}{3}\Big[1 - \frac{U^2}{15} + \ldots\Big] \quad (U \to 0), \qquad \to \frac{\pi}{2U} \quad (U \to \infty). \tag{30}$$

Because $\mathrm{Si}(U) > |j_1(U)|$ for all $U > 0$, this profile has *no zeros* and a non-oscillatory $1/\rho$ tail. Its intensity half width at half maximum is 1.80× that of the single-frequency profile $j_0 - j_1/U$. The 1D features (faster side-lobe decay, first zero doubled) are therefore specific to the 1D line. [numerical check: closed form vs. quadrature 10^{-14}; minimum over $0 < U \le 200$ is $+5.9 \times 10^{-3}$]

Dependence on the normalization. Physical energy normalization changes the optimal weights. On the 2D line the per-frequency cost is $\int (k_z/k)|c|^2\mathrm{d}k_x$, so $c \propto k/k_z$ and the $t = 0$ profile becomes $\int_0^{\omega_m} \omega J_0(\omega x/c)\mathrm{d}\omega \propto J_1(U)/U$ (versus single-frequency $J_0(U)$; first-zero ratio 1.593). In 3D the incoming energy of a Herglotz pulse is $\propto \int \mathrm{d}\omega \int \mathrm{d}\Omega|\mathbf{a}|^2/\omega^2$, so $\mathbf{a} \propto \omega^2$ and the profile is $\propto [2\sin U - U\cos U - \mathrm{Si}(U)]/U^3$, which does have zeros (first at $U = 3.797$, versus 2.744 for a single frequency). The qualitative conclusion, that temporal localization costs spatial localization, survives. The factor of 2 and the side-lobe behavior do not.

7 Focusing in complex media (tutorial Sec. 3.2)

7.1 The bound and the optimal currents

Currents s on an aperture A produce $\psi(\mathbf{x}_0) = \Gamma_{0A} s$. For a target polarization ϕ, the objective $|\phi^\dagger \Gamma_{0A} s|^2$ with $\|s\| = 1$ is rank one, so by (4)

$$\max |\phi^\dagger \psi(\mathbf{x}_0)|^2 = \|\Gamma_{0A}^\dagger \phi\|^2 = \int_A \left|(\Gamma_{0A}^\dagger \phi)(\mathbf{x}')\right|^2 \mathrm{d}\mathbf{x}', \qquad s_{\mathrm{opt}} \propto \Gamma_{0A}^\dagger \phi. \tag{31}$$

Nothing about the medium has been assumed. If the medium is reciprocal, $\Gamma(\mathbf{x}_0, \mathbf{x}')^{\mathrm{T}} = \Gamma(\mathbf{x}', \mathbf{x}_0)$, then $(\Gamma_{0A}^\dagger \phi)(\mathbf{x}') = [\Gamma(\mathbf{x}', \mathbf{x}_0)\phi^*]^*$, the conjugate of the field on A radiated by a dipole ϕ^* at $\mathbf{x}_0$. If the medium is not reciprocal, generalized reciprocity $\Gamma_\nu(\mathbf{x}, \mathbf{x}')^{\mathrm{T}} = \Gamma_{\nu^{\mathrm{T}}}(\mathbf{x}', \mathbf{x})$ gives the same recipe in the *transposed* medium.

Far-field aperture and the LDOS. For a closed aperture in the far zone of a lossless medium, $|\mathbf{E}| = Z_0|\mathbf{H}|$ and $\int_A |\mathbf{E}_0|^2 \mathrm{d}A = 2Z_0 P_{\mathrm{rad}}$, where $P_{\mathrm{rad}} = \frac{\omega}{2}\operatorname{Im}(\mathbf{p}^* \cdot \mathbf{E}(\mathbf{x}_0))$ is the power radiated by the dipole, i.e. the LDOS. For apertures in the near field the bound is larger.

7.2 A point-dipole medium with controllable symmetries

To test these statements we use electric point scatterers $\mathbf{p}_j = \alpha_j \mathbf{E}_{\mathrm{loc}}(\mathbf{x}_j)$ with $\alpha_j^{-1} = \alpha_{0,j}^{-1} - i\frac{k^3}{6\pi\varepsilon_0}\mathbb{I}$, where the radiation-reaction term is $\varepsilon_0^{-1} k^2 \operatorname{Im} \mathsf{G}_0(0)$. With $\mathbf{E} = \varepsilon_0^{-1} k^2 \mathsf{G}_0 \mathbf{p}$, the medium Green's function is

$$\Gamma(\mathbf{x}, \mathbf{x}') = \frac{k^2}{\varepsilon_0}\Big[\mathsf{G}_0(\mathbf{x}, \mathbf{x}') + \frac{k^2}{\varepsilon_0}\mathsf{G}_0(\mathbf{x}, S)\,\mathsf{T}\,\mathsf{G}_0(S, \mathbf{x}')\Big], \qquad \mathsf{T} = \big(\operatorname{diag}\alpha_j^{-1} - \tfrac{k^2}{\varepsilon_0}\mathsf{G}_{SS}\big)^{-1}, \tag{32}$$

where G_{SS} couples distinct scatterers. All blocks of G_0 are symmetric and $\mathsf{G}_0(\mathbf{x}, \mathbf{x}') = \mathsf{G}_0(\mathbf{x}', \mathbf{x})$, so $\mathsf{T}^{\mathrm{T}} = \mathsf{T}|_{\alpha \to \alpha^{\mathrm{T}}}$. The symmetries of the medium are then controlled entirely by α_0:

- *reciprocal* $\Leftrightarrow \alpha_0^{\mathrm{T}} = \alpha_0$. In general, $\Gamma_\alpha(\mathbf{x}, \mathbf{x}')^{\mathrm{T}} = \Gamma_{\alpha^{\mathrm{T}}}(\mathbf{x}', \mathbf{x})$.
- *lossless* $\Leftrightarrow \alpha_0^\dagger = \alpha_0$. The power absorbed by dipole j is $\frac{\omega}{2}\mathbf{p}^\dagger[(\alpha_0^{-1})^\dagger - \alpha_0^{-1}]\mathbf{p}/(2i)$, the radiation-reaction term accounting exactly for the scattered power $\frac{\omega k^3}{12\pi\varepsilon_0}|\mathbf{p}|^2$.
- *time-reversal invariant* $\Leftrightarrow \alpha_0$ real. Under $(\mathbf{E}, \mathbf{H}, \mathbf{p}, \mathbf{m}) \to (\mathbf{E}^*, -\mathbf{H}^*, \mathbf{p}^*, -\mathbf{m}^*)$ the constitutive relation $\mathbf{p} = \alpha_0 \mathbf{E}_{\mathrm{loc}}$ maps to itself iff $\alpha_0^* = \alpha_0$. The radiation-reaction term is part of the outgoing boundary condition, not of the medium.

The four test media are lossless reciprocal (α_0 real symmetric), lossy reciprocal (complex symmetric), *TRI but non-reciprocal* (real, non-symmetric), and gyrotropic (Hermitian, non-symmetric: lossless, non-reciprocal, non-TRI).

7.3 Numerical results

Table 2 tests the bound and the recipes with random clusters of 16 scatterers in a λ^3 box, a random complex target polarization, and electric currents on either an open near-field patch or a closed sphere of radius 6λ. Γ is built directly from T, without invoking reciprocity. The iterative eigensolution of the full operator confirms the rank-one maximum in all cases (rel. difference $\leq 2\times10^{-15}$).

For the LDOS connection, closed spheres about $\mathbf{x}_0$ in the lossless reciprocal medium give bound/$(2P_{\mathrm{rad}})$ = 94.8, 45.8, 1.012 and 1.00012 for R = 0.06, 0.3, 3 and 30λ. The sphere with $R = 0.3\lambda$ passes through the cluster, and any aperture is allowed.

Table 2: Achieved focal intensity divided by the bound, for currents prescribed by the conjugated dipole field computed in the actual medium ("recipe") or in the transposed medium (`complex_media_focusing.py`).

	open near-field patch		closed sphere, $R = 6\lambda$	
medium	recipe	transposed	recipe	transposed
lossless, reciprocal	1.00000000	1.00000000	1.00000000	1.00000000
lossy, reciprocal	1.00000000	1.00000000	1.00000000	1.00000000
TRI, non-reciprocal	0.216	1.00000000	0.084	1.00000000
gyrotropic	0.265	1.00000000	0.231	1.00000000

7.4 Partial time reversal on a closed surface

In the representation $\psi = (\mathbf{E}, i\mathbf{H})$, $s = (i\mathbf{J}_e, -\mathbf{J}_m)$ of the tutorial, complex conjugation is time reversal: $\mathbf{E} \to \mathbf{E}^*$, $\mathbf{H} \to -\mathbf{H}^*$, $\mathbf{J}_e \to -\mathbf{J}_e^*$, $\mathbf{J}_m \to \mathbf{J}_m^*$ (for point dipoles, $\mathbf{p} \to \mathbf{p}^*$, $\mathbf{m} \to -\mathbf{m}^*$). The four-step construction of the tutorial proceeds as follows. The Love currents $\mathbf{J}_s = \hat{\mathbf{n}} \times \mathbf{H}$, $\mathbf{J}_{m,s} = -\hat{\mathbf{n}} \times \mathbf{E}$ on a closed surface S enclosing all sources and scatterers radiate $(\mathbf{E}, \mathbf{H})$ outside S and zero inside, in the actual medium, since the scatterers inside S see no field. Their negatives are the perfectly cancelling currents s_{pcc}. If the medium is TRI, conjugation gives $M\psi^* = s_0^* + s_{\mathrm{pcc}}^*$, so

$$\psi_{\mathrm{TR}} \equiv \Gamma s_{\mathrm{pcc}}^* = \psi^* - \Gamma s_0^* = \begin{cases} -2i \operatorname{Im} \psi & \mathbf{x} \text{ inside } S \ (s_0 \text{ real}), \\ -\Gamma s_0^* & \mathbf{x} \text{ outside } S. \end{cases} \tag{33}$$

The entrywise-$\operatorname{Im}\Gamma$ form of the tutorial therefore holds for field points inside S. Reciprocity was not used at any step.

The numerical test uses a real dipole inside a sphere S of radius λ, with one scatterer placed 0.05λ from S to test the claim that no far-field assumption is needed. Fields and Love currents are evaluated with a 180×360 Gauss–Legendre×trapezoid quadrature on S. Time-reversed currents are radiated in the medium, and the result is compared with $-2i \operatorname{Im} \mathbf{E}$ at 25 interior points:

medium	$\max \lvert\psi_{\mathrm{TR}} + 2i \operatorname{Im} \mathbf{E}\rvert / \max \lvert 2 \operatorname{Im} \mathbf{E}\rvert$ inside S	extinction-theorem residual
lossless, reciprocal (TRI)	7×10^{-8}	2×10^{-7}
TRI, non-reciprocal	1×10^{-6}	4×10^{-6}
lossy, reciprocal (not TRI)	0.45	2×10^{-7}
gyrotropic (not TRI)	1.24	1×10^{-6}

The residuals of the TRI cases fall with quadrature refinement, tracking the extinction-theorem residual (at 120×240 they are 5×10^{-6} and 6×10^{-5}). The failures are $O(1)$ and do not change. This supports two statements of the tutorial: that time reversal, not reciprocity, underlies the conjugate-field identity, and that the time-reversal recipe and the reciprocity recipe are distinct. In the TRI non-reciprocal medium the time-reversed field is exactly $-2i \operatorname{Im} \mathbf{E}$, yet the conjugate-field recipe reaches only 8–22% of the maximum focal intensity (Table 2). Figure 5 shows the comparison on a slice through the cluster.

7.5 Space-time version

With sources $s(\mathbf{x}, t)$ on A for $t \in [0, T]$, $\psi(\mathbf{x}_0, t_0) = \int_0^T \mathrm{d}t \int_A \Gamma_{\mathrm{ret}}(\mathbf{x}_0, t_0; \mathbf{x}, t) s(\mathbf{x}, t)$, which is again rank one. By time-domain reciprocity and time-translation invariance, $\Gamma_{\mathrm{ret}}(\mathbf{x}_0, t_0; \mathbf{x}, t)^{\mathrm{T}} =$

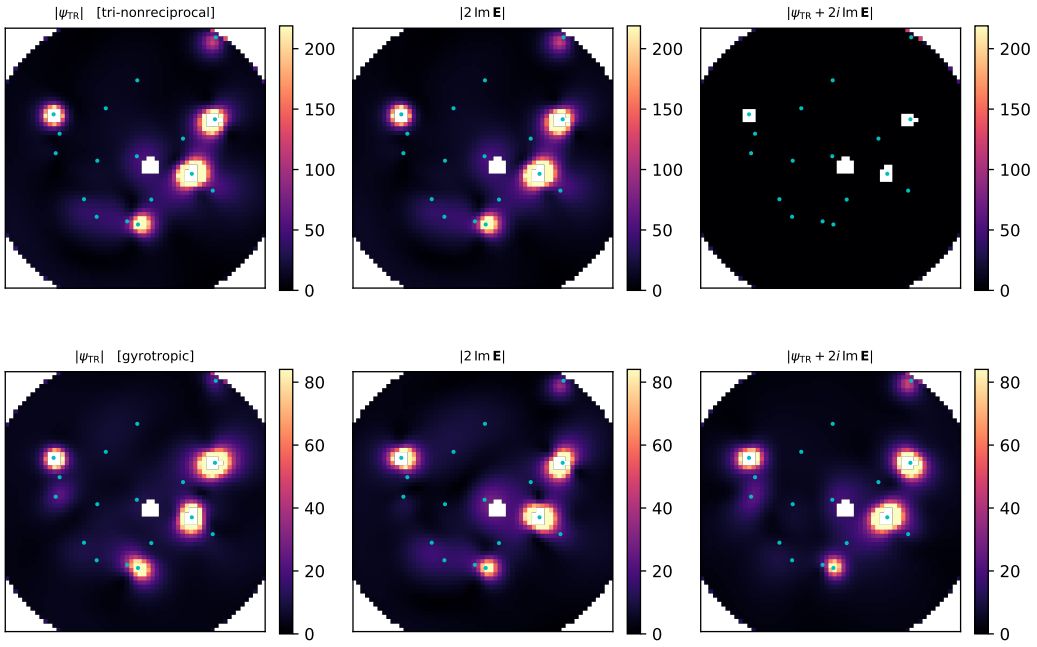


Figure 5: Partial time reversal on a slice through the scatterer cluster (dots: scatterer positions projected onto the slice; +: source; masked: points within 0.06λ of a scatterer or the source). Top: TRI but non-reciprocal medium, where the time-reversed field equals $-2i\,\mathrm{Im}\,\mathbf{E}$ everywhere inside S. Bottom: gyrotropic medium (not TRI), where it does not.

$\Gamma_{\mathrm{ret}}(\mathbf{x}, t_0 - t; \mathbf{x}_0, 0)$. The optimal source is then $s(\mathbf{x}, t) \propto \psi_0(\mathbf{x}, t_0 - t)$: the field of a dipole fired at $(\mathbf{x}_0, 0)$, time-reversed. The bound is

$$\max |\phi^{\mathrm{T}} \psi(\mathbf{x}_0, t_0)|^2 = \int_{t_0 - T}^{t_0} \mathrm{d}\tau \int_A |\psi_0(\mathbf{x}, \tau)|^2 \mathrm{d}\mathbf{x}, \tag{34}$$

which reduces to the window $[0, T]$ written in the tutorial when the focal time is the end of the source window, $t_0 = T$.

8 Index of numerical checks

result	script (repository)	agreement
plane-wave power formula	`power_normalization_check.py`	10^{-16}
focal-intensity maxima, optimal polarization	`focal_intensity_bounds.py`	3×10^{-13}
amplitude-optimal beam per physical power	`focal_intensity_bounds.py`	10^{-14}
focal-plane fields, closed forms, $\frac{1}{2}$ of 4π	`focal_plane_fields.py`	2×10^{-14}
widths (Table 1)	`focal_plane_fields.py`	10^{-4} (root finding)
VSW amplitudes, P_{in}, $k^2/3\pi c$, $\mathrm{Im}\,\mathsf{G}_0$	`fourpi_spherical_waves.py`	10^{-10}–10^{-15}
super-oscillation penalties, $S_{\max}$	`superoscillation.py`	10^{-8}
space-time profiles	`spacetime_focusing.py`	10^{-14}
complex-media bound, recipes, TR identity	`complex_media_focusing.py`	10^{-15} / 10^{-7}

All scripts run in seconds on a laptop (Python 3 with NumPy, SciPy, Matplotlib); `pytest tests/` runs the regression versions.